

LLM Landscape 2025

LLM Main Track

Transformer based models

Deepseek V3/R1

GLM

Qwen3-Next

Trend: Bigger models with ↓ inference
req

→ Kimi - 1T params

→ Can't run at home but possibly
in datacenter

Tricks: GQA

Grouped Query Attention

GQA is not new

Not new. Proposed in 2023

Gemma 3

Mistral 3.1

Llama 4

K2

gpt-oss

GLM

} Not
comprehensive

All use GQA

GQA replaces MHA

Sits inside Transformer block

GQA reduces KV cache size by
sharing K/V across multiple Q heads

$$\text{group size} = n\text{-heads} / n\text{-kv-groups}$$

$$= 6 / 3$$

$$= 2$$

KV cache being small is important

If we make only one group,
all queries will share the same
set of K/V values. This is
EXTREME.

This is called Multi Query Attention
(MQA).

Accuracy of model reduces.

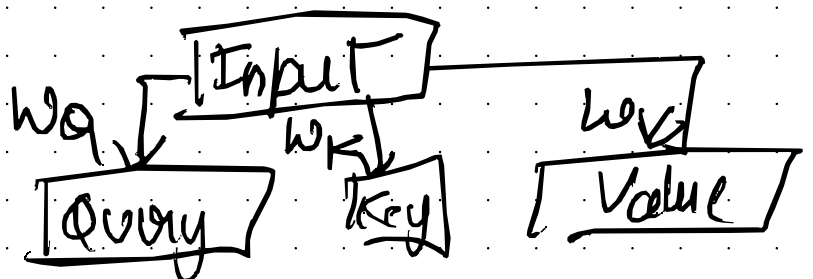
This is a popular trick.

Multi Head Latent Attention (MLA)

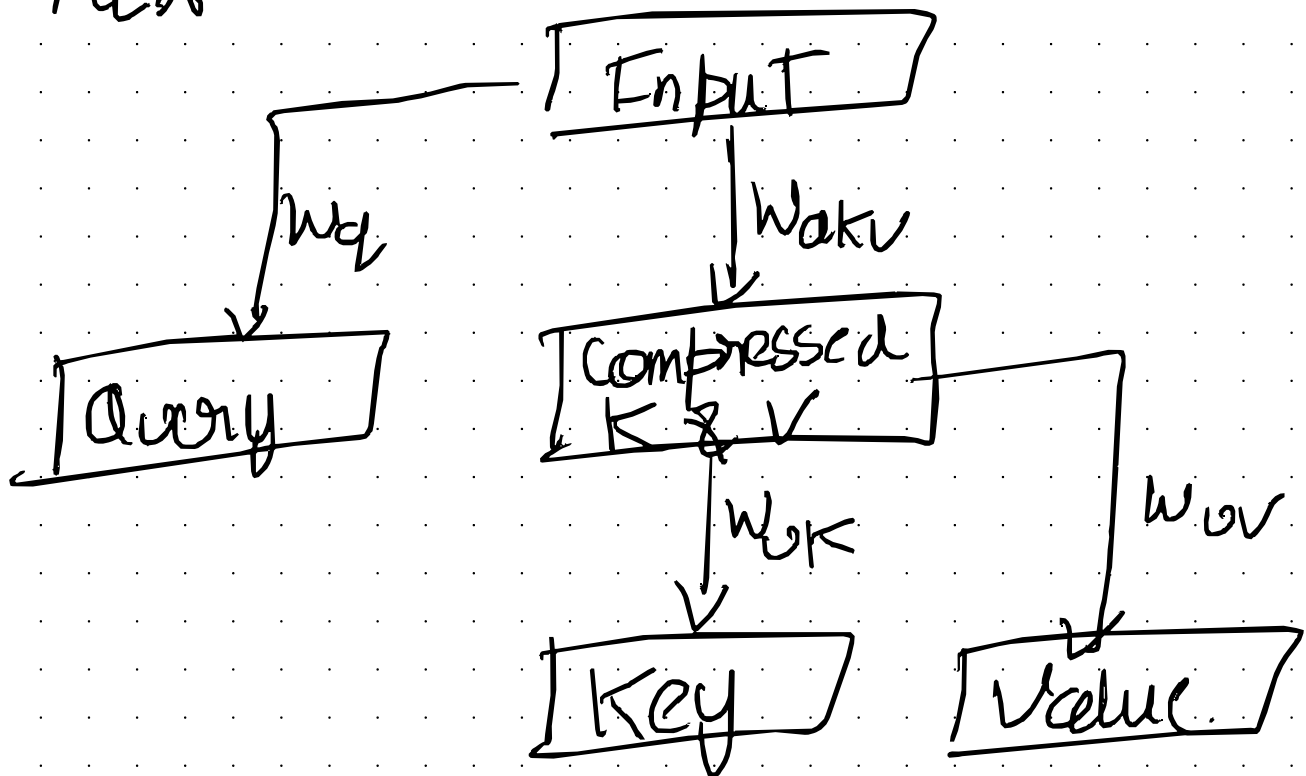
DS $R3/V1$ & $K2$.

Similar to GQA, but diff implementation.
can be combined by GQA.

Regular MHA :



MLA



Instead of directly going to K & V ,
we first go to a compressed state.

Saving space by compressing.

During inference, we 'up-project'
the latent representation to a

It's kinda like LORA

Projection that down & up sample.
Saves vs memory.

Performs better than GDA in terms of modelling performance

Sliding Window Attention (SWA)

- Not super popular anymore
- Gemma-3, Mistral-3.1, gpt-oss?

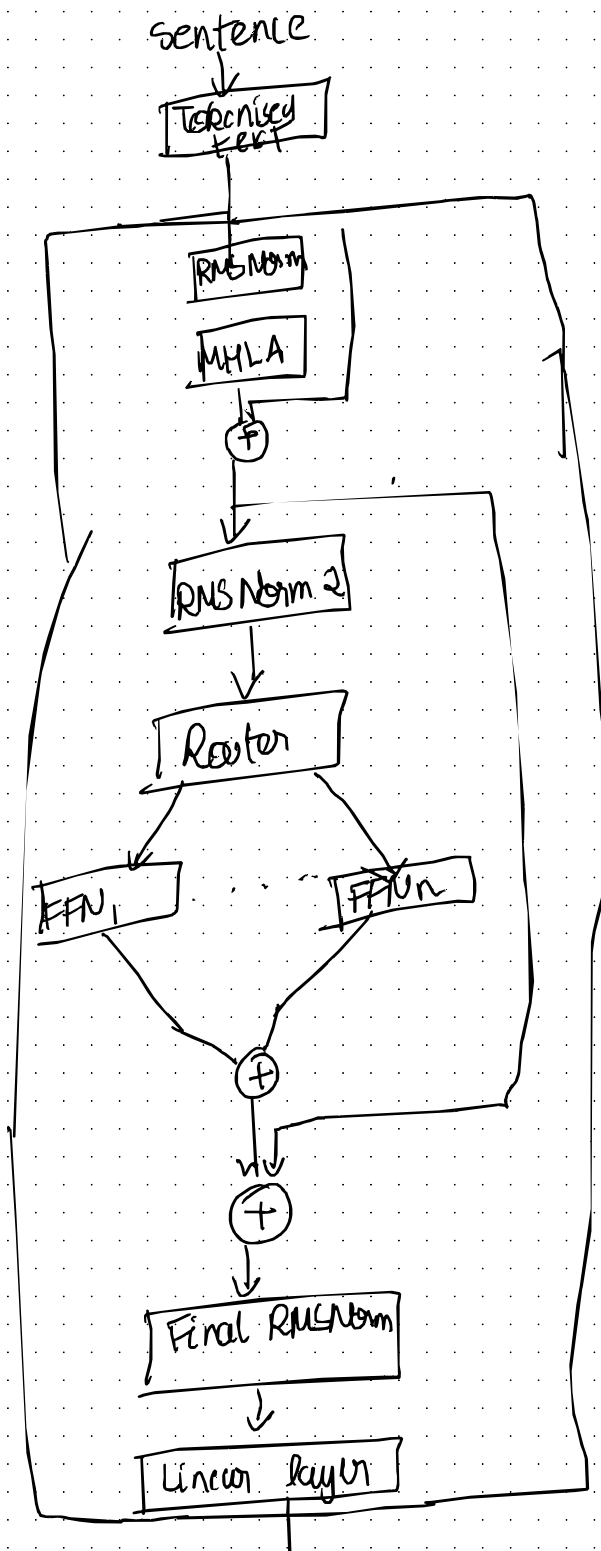
Regular attention: every token attends to every other token before itself

SWA limits how far back a token can attend to. On long ctx, this will save compute & memory.

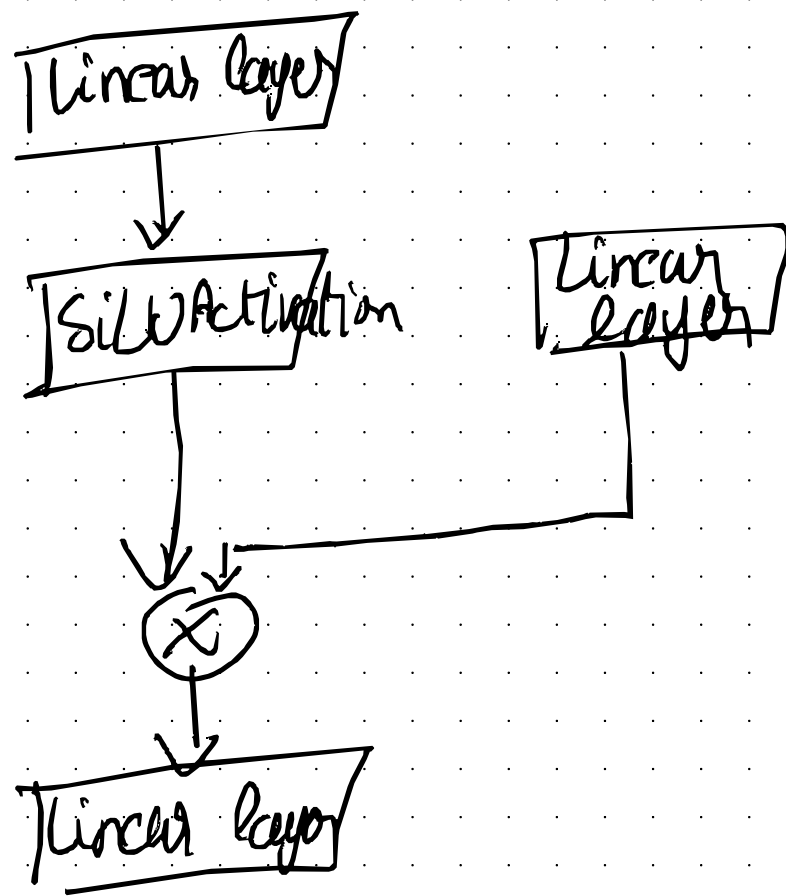
Choosing it with 5:1 ratio, according to Gemma3 paper, it doesn't impact modelling performance.

Mixture of Experts (MoE)

- Very common now
- Used by everyone
- MoE has a router, which routes to specific FFN layers
- MoE AFN are generally applied only in the feedforward layers.



Feed forward (SwiGLU Model)



V3 was the first model to introduce 'sparse MOE'. Total experts are 256.

Shared expert: Always activated
g experts: Activated based on input

This ratio of active to total expert ($8:256 \rightarrow 1:32$) was the highest at the time.

Usual ratios were around $1:8$

Saves a lot of params in inference.

MoE has all params in training, but in inference we use only a subset

Alternatives to Transformers

Minimax M1, Open 3-next, DS V3-2-exp

→ Same decoder style transformer

→ Cuts FLOP/KV memory at long context

Cons → Added complexity

→ Trades accuracy for efficiency.

Small Recursive Transformers

Hierarchical Reasoning Model

↙
Mixture of
recursions

↘
tiny Reasoning
Models

→ very small arch

→ Good generalization on puzzles

Cons

→ special purpose

→ limited to puzzles.

Code World Model

→ Model simulates how code is
executed. Verifiable intermediate
stages

→ Adds complexity

Text Diffusion Models

- Iterative denoising is a new idea for text
- Better parallelism (no next-token dependence)

Cons

- Can't stream answers
- Doesn't benefit from CoT
- NOT SOTA

Also saw X posts showing that diffusion text models are not doing great.

Models: DiffuSe4
LLaDA
Dream 7B

In ex of reasoning agents, not sure how these fit

Liquid Foundation Models (LFM)

- LFM 1
- LFM MOE
- Fresh continuous time state updates
& differential equation approach
- Parameters efficient

Cons

- Doesn't benefit from CoT
- Solid models but not SOTA

Don't know much, but look parameter efficient.

Transformer RNN Hybrids

→ RWKV & RWKV-1

→ Doesn't need KV cache (constant memory)

→ Efficient on long Ltx tasks, but has poor performance.

→ It is an intermediate step, like an RNN, and doesn't get more expensive as the context increases

→ But, because everything is latent, you compress everything to same spec, and end up losing information.

State Space Models

- Mamba
- S^h being used for voice models (Caster) $\rightarrow$
- Pretty popular
- Hunyuan T1 model is also an SSM.

Transformer SSM Hybrid

- SSM based models had too many limitations, so hybrids are now commonly used.
- Can't escape the $O(n^2)$ penalty!!
- Hybrids are Jamba, Samba & Hunyuan T1

LSTM Models

→ x LSTM(2024)

→ Efficient models for devices.